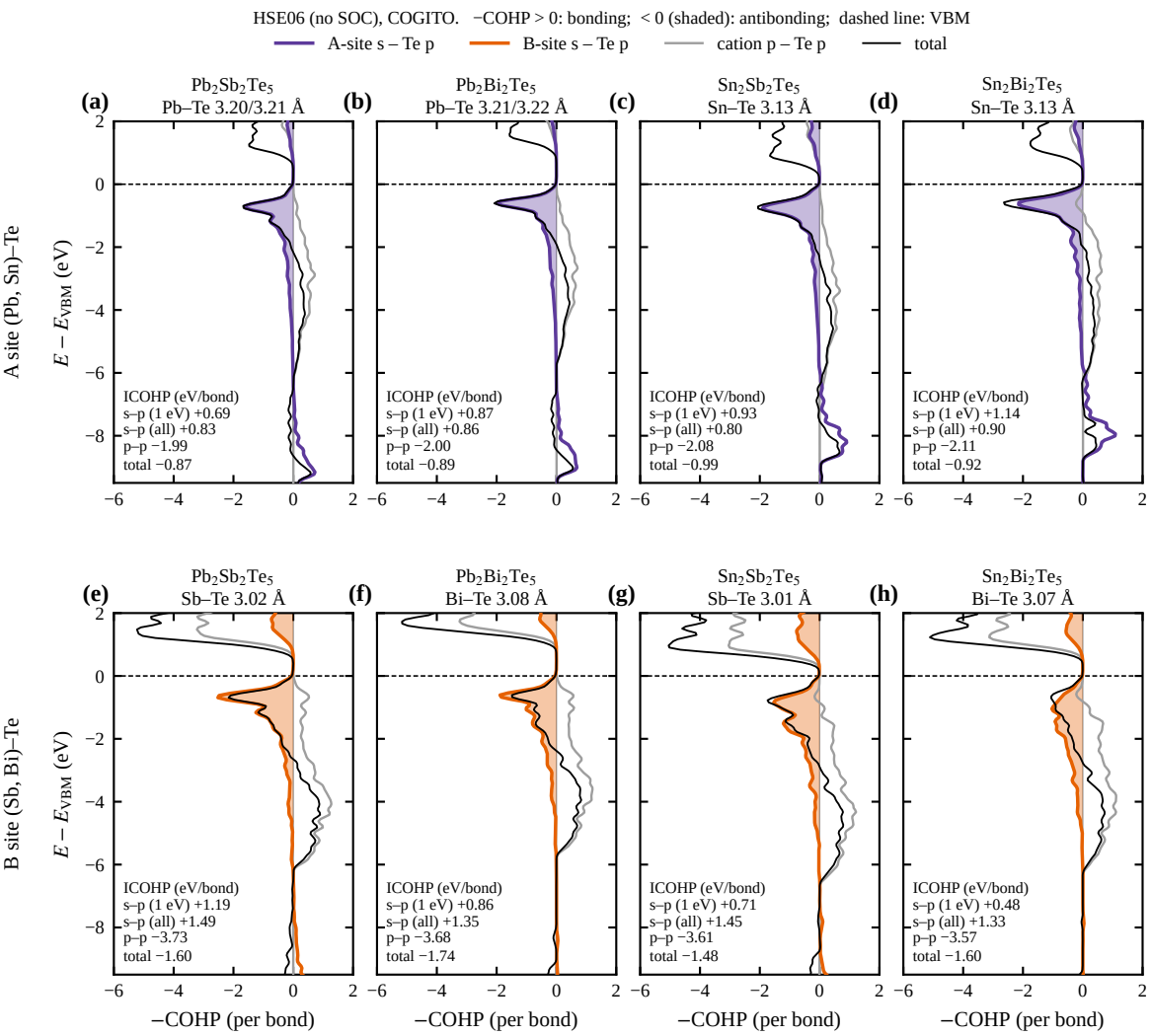

HSE06 (no SOC), COGITO. −COHP > 0: bonding; < 0 (shaded): antibonding; dashed line: VBM
A-site s – Te p
B-site s – Te p
cation p – Te p
total
(a)
Pb2Sb2Te5
Pb–Te 3.20/3.21 Å
ICOHP (eV/bond)
s–p (1 eV) +0.69
s–p (all) +0.83
p–p −1.99
total −0.87
(b)
Pb2Bi2Te5
Pb–Te 3.21/3.22 Å
ICOHP (eV/bond)
s–p (1 eV) +0.87
s–p (all) +0.86
p–p −2.00
total −0.89
(c)
Sn2Sb2Te5
Sn–Te 3.13 Å
ICOHP (eV/bond)
s–p (1 eV) +0.93
s–p (all) +0.80
p–p −2.08
total −0.99
(d)
Sn2Bi2Te5
Sn–Te 3.13 Å
ICOHP (eV/bond)
s–p (1 eV) +1.14
s–p (all) +0.90
p–p −2.11
total −0.92
A site (Pb, Sn)–Te
B site (Sb, Bi)–Te
E − EVBM (eV)
(e)
Pb2Sb2Te5
Sb–Te 3.02 Å
ICOHP (eV/bond)
s–p (1 eV) +1.19
s–p (all) +1.49
p–p −3.73
total −1.60
(f)
Pb2Bi2Te5
Bi–Te 3.08 Å
ICOHP (eV/bond)
s–p (1 eV) +0.86
s–p (all) +1.35
p–p −3.68
total −1.74
(g)
Sn2Sb2Te5
Sb–Te 3.01 Å
ICOHP (eV/bond)
s–p (1 eV) +0.71
s–p (all) +1.45
p–p −3.61
total −1.48
(h)
Sn2Bi2Te5
Bi–Te 3.07 Å
ICOHP (eV/bond)
s–p (1 eV) +0.48
s–p (all) +1.33
p–p −3.57
total −1.60
−COHP (per bond)